

Multiple Choice (10 marks)

1. If the speed of an airplane is measured by means of a stagnation tube, what correction must be applied to the constant-density indication when the plane is cruising at 300 mph?
 - (a) - 10 mph
 - (b) 25 mph
 - (c) -30 mph
 - (d) -15 mph
 - (e) - 5 mph
2. The field winding of a shunt motor has a resistance of 110 ohms, and the voltage applied to it is 220 volts. What is the amount of power expended in the field excitation ?
 - (a) 660 watts
 - (b) 880 watts
 - (c) 200 watts
 - (d) 120 watts
 - (e) 110 watts
3. In an experiment involving the toss of two dice, what is the probability that the sum is 6 or 7?
 - (a) $5/36$
 - (b) $3/12$
 - (c) $6/36$
 - (d) $1/6$
 - (e) $17/36$
4. What is the probability of throwing a "six" with a single die?
 - (a) $3/6$
 - (b) $1/12$
 - (c) $1/6$
 - (d) $2/6$
 - (e) $1/8$
5. Find the magnitude of E for a plane wave in free space, if the magnitude of H for this wave is 5 amp/meter.

- (a) 1200 volts/meter
- (b) 5 ohms
- (c) 1883 volts/meter
- (d) 940.5 volts/meter
- (e) 5 volts/meter

True / False (10 marks)

Answer True or False

6. True or False: The impedance of a lossless medium with $\mu_r = 10$ and $\epsilon_r = 5$ is 266 ohms.
7. True or False: A combinational PLD with a fixed AND array and a programmable OR array is referred to as a PAL.
8. True or False: The lowest critical frequency of a network occurs at zero for an RL circuit, indicating a short circuit.
9. True or False: The electric field produced by a varying magnetic field $B = B_0 e^{j\omega t}$ is given by $E = -\frac{1}{\omega} \frac{dB}{dt}$.
10. True or False: The partial-fraction expansion of $F(s) = (8s + 2) / (s^2 + 3s + 2)$ is $F(s) = [(-2) / (s + 1)] + [(10) / (s + 2)]$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the quantization process of a signal and derive the mean-square quantization error, considering the error uniformly distributed in the range $-S/2$ to $+S/2$.
12. Given a sphere of radius a with a uniform charge density ρ inside and zero outside, derive the electric potential throughout the entire space by solving Poisson's equation. Discuss the implications of the result.

End of Paper